

AI-Powered Digital Marketplace for Upcycling Sri Lanka's Textile Waste

Project ID: R26-IT-081 | Project Proposal Presentation

BSc (Hons) in Information Technology

Sri Lanka Institute of Information Technology



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Our Team

Meet the dedicated individuals driving this innovative project:



Sandeepa S.A.C.

AI Fabric Classification Module



Siriwardhana N.S.T.N.

Environmental Impact Prediction Module



Madushanka H.M.T.G.

Traceability, Recommendation & AI Pricing Module



Thalagala T.A.D.V.

Marketplace Platform & Predictive Analytics Module

Introduction

Sri Lanka's **textile and apparel industry** generates a large amount of fabric waste during:

- Fabric cutting
- Production processes
- Quality control operations

Due to the absence of a **transparent and scalable redistribution system**, much of this textile waste is:

- Burned by manufacturers
- Disposed of in landfills

This leads to:

- Environmental pollution
- Increased carbon emissions
- Loss of reusable textile resources

At the same time, **small-scale designers, artisans, and SMEs struggle to access affordable raw materials for sustainable production.**





The Research Problem

Current textile waste management systems have several limitations:

- Manual sorting is **inconsistent, slow, and subjective**
- No **automated classification** of textile waste
- No **AI-based pricing or recommendation systems**
- Lack of **traceability mechanisms**
- No **environmental impact tracking**

As a result:

- Manufacturers lose potential revenue
- Buyers cannot find a reliable waste supply
- Environmental damage increases

Statistical Evidence

- Global textile waste: **92 million tons annually**
- Less than **15% is recycled**
- Sri Lanka's apparel industry contributes **~7% to GDP**, but produces significant unused textile waste



Research Objectives

Main Objective

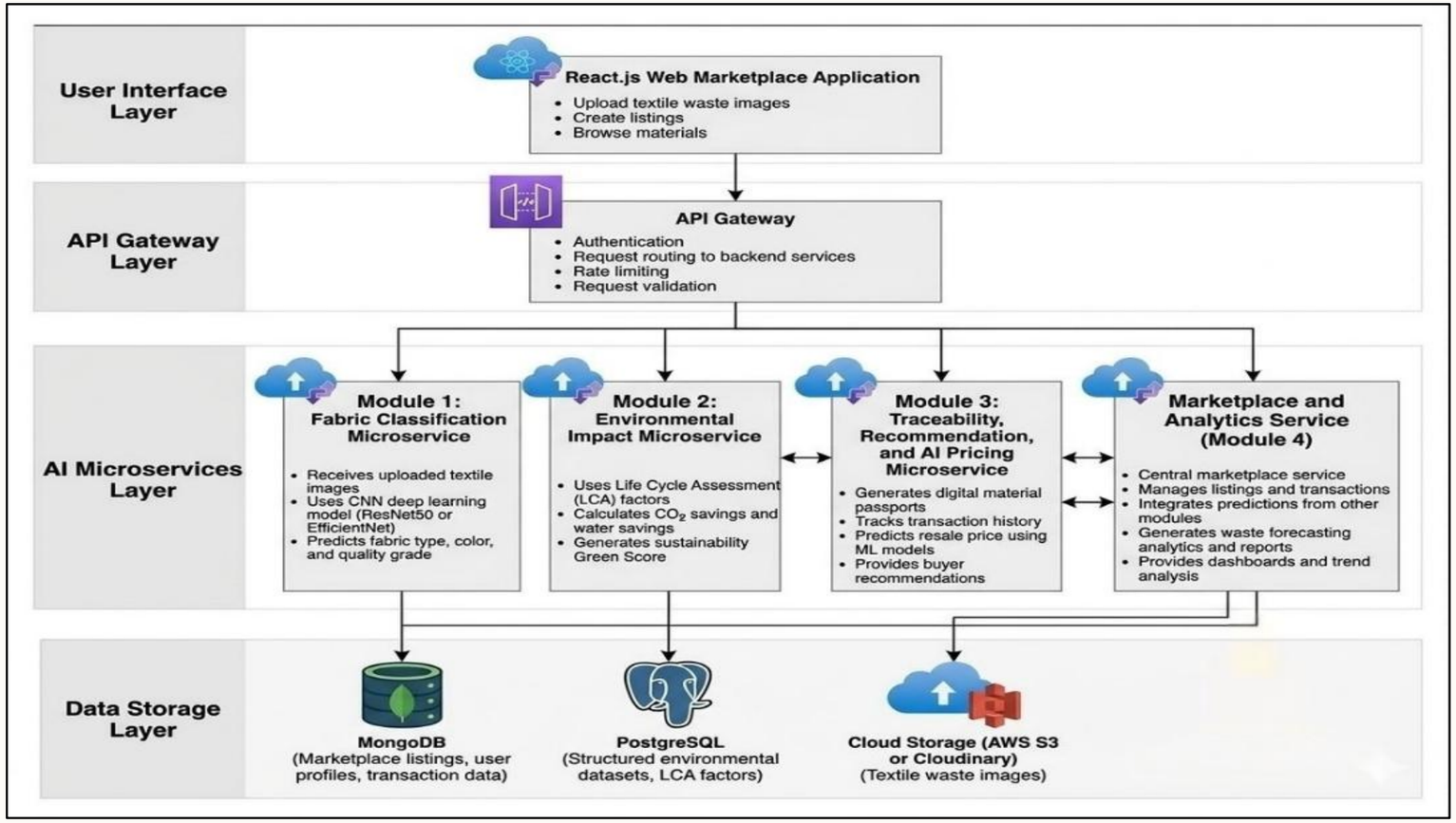
Develop and evaluate an **AI-powered digital marketplace** that enables intelligent classification, valuation, environmental impact analysis, and redistribution of textile waste in Sri Lanka, promoting circular economy **principles**.

Specific Objectives

1. Module 1 – AI Fabric Classification (Sandeepa)
2. Module 2 – Environmental Impact Prediction - (Siriwardhana)
3. Module 3 – Traceability, Recommendations & AI Pricing - (Madushanka)
4. Module 4 – Marketplace Platform & Predictive Analytics - (Thalagala)

System Overview

Our proposed system architecture integrates AI-powered modules into a seamless digital marketplace, driving efficiency and sustainability in textile waste management.



All core modules communicate efficiently through FastAPI microservices, ensuring modularity, scalability, and robust performance across the entire platform.

Individual Component

AI-Driven Fabric Classification and Cataloguing Module

Student: **Sandeepa S.A.C**

Student ID: **IT22281814**

Project ID: **R26-IT-081**



Background:

Most textile factories rely on **manual visual inspection** to sort fabric waste based on:

- Fabric type
- Colour
- Quality grade

This process is:

- Time-consuming
- Inconsistent
- Highly subjective



Computer vision, utilizing Convolutional Neural Networks (CNNs), can automatically recognize fabric characteristics even in **crumpled or folded waste materials.**

Research Problem

1 Manual textile-waste sorting has several challenges:

- Time-consuming
- Subjective human judgement
- High classification error rate

2 Additionally,

Many existing AI models are trained on **clean laboratory fabric samples**, which perform poorly on **real-world post-industrial textile waste**.



Research Gap

Our project directly addresses critical limitations in current textile waste management by leveraging advanced AI techniques tailored for real-world scenarios.

Feature	Existing Solutions	Proposed Module
Waste Classification	Manual / expensive hyperspectral methods	CNN with transfer learning
Dataset	Clean lab images	5,000+ real textile-waste images
Real-World Performance	Low accuracy on crumpled samples	Target >85% accuracy



Research Objectives

1

Main Objective

Develop a **deep learning model for automated textile waste classification.**

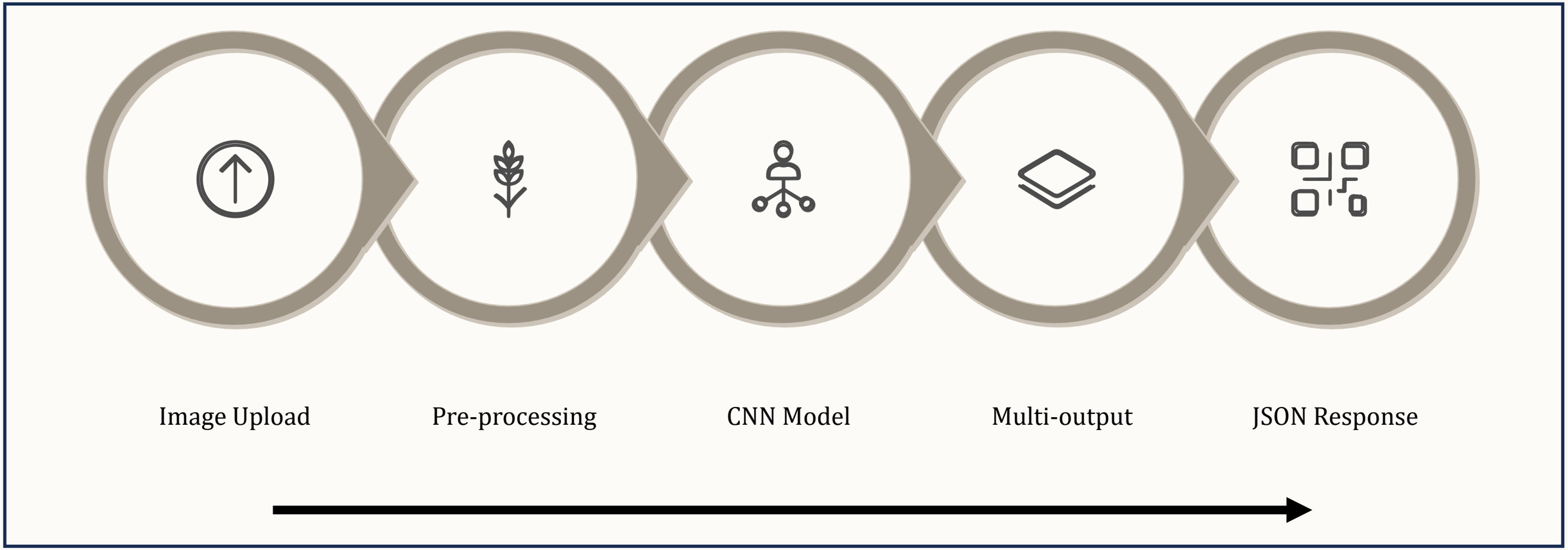
2

Sub Objectives

- Train CNN models using **transfer learning**
- Achieve **>85% accuracy**
- Deploy the model as a **FastAPI microservice**

System Architecture

Our module transforms raw textile waste images into structured, classified data, forming the backbone of the digital marketplace.



This architecture ensures robust and accurate classification, providing essential data for pricing, recommendations, and environmental impact analysis.

Tools and Technologies

Our project leverages a robust stack of cutting-edge tools and frameworks to ensure high performance, scalability, and efficiency across all modules.

Programming Language

Python 3.9+: The primary language for all development, chosen for its extensive libraries and AI capabilities.

Deep Learning Frameworks

TensorFlow / Keras: Industry-standard frameworks for building, training, and optimizing complex neural networks.

Model Techniques

Transfer Learning: Utilizing pre-trained models like **ResNet-50** and **EfficientNetB3** for accelerated training and superior accuracy on textile data.

Deployment

FastAPI & Docker: Powering high-performance microservices and ensuring containerized, portable deployments for seamless integration.

Cloud Computing

Google Colab Pro / AWS GPU: Harnessing powerful GPU resources for intensive model training, experimentation, and scalable operations.

Software Requirements

1

Functional Requirements

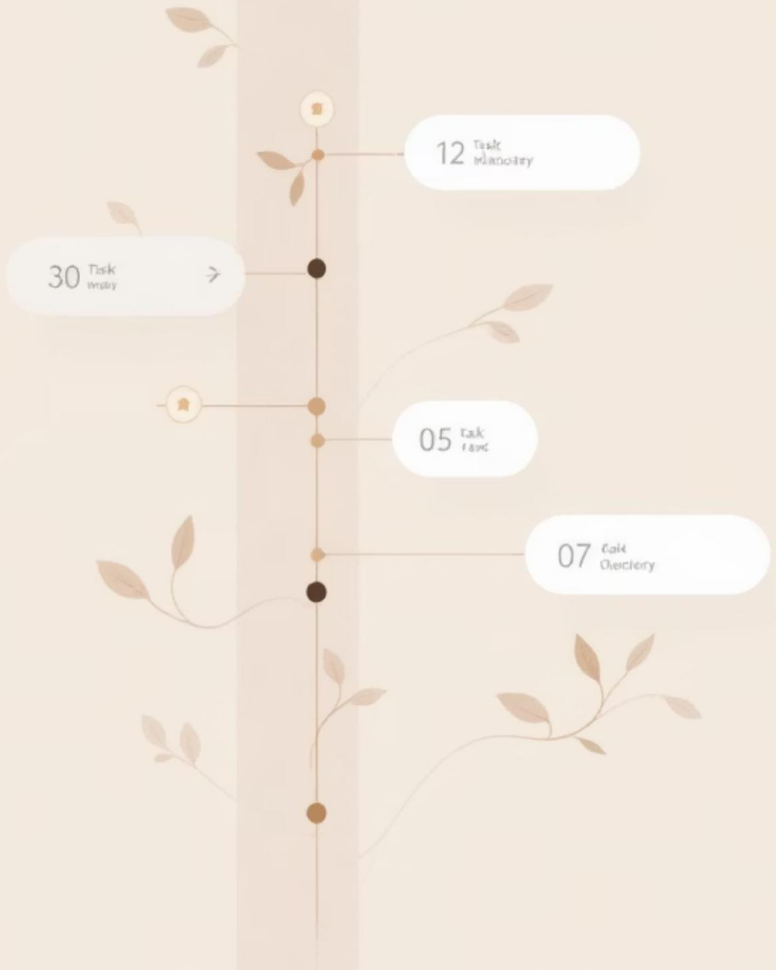
- Accept textile waste images (JPG/PNG format)
- Perform **automatic fabric classification**
- Predict **fabric type, colour, and quality grade**
- Return prediction results with **confidence scores via API**

2

Non-Functional Requirements

- Classification accuracy **greater than 85%**
- API response time **less than 2 seconds**
- Robust to **lighting, background, and angle variations**
- Scalable deployment using **Docker and FastAPI**





Work Breakdown Structure

Our project's systematic approach ensures a clear roadmap from research to deployment for the AI Fabric Classification module.

1 **Literature Review**
Comprehensive study of existing AI models and textile waste management techniques.

2 **Data Collection & Labeling**
Gathering and annotating a diverse dataset of Sri Lankan textile waste images.

3 **Data Augmentation**
Expanding the dataset through various techniques to improve model robustness.

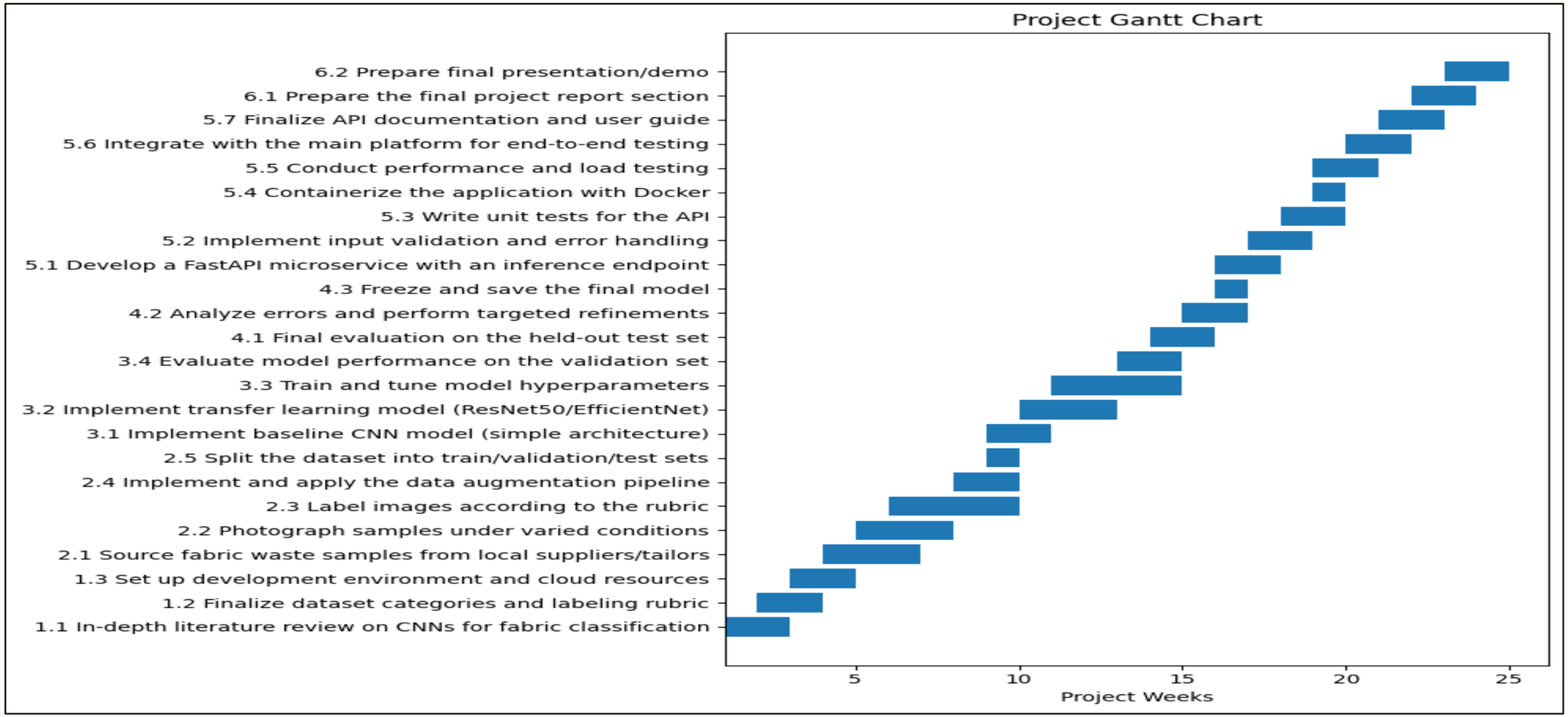
4 **CNN Baseline Model Development**
Building an initial Convolutional Neural Network model for classification.

5 **Transfer Learning & Fine-tuning**
Adapting pre-trained models (ResNet-50, EfficientNetB3) for enhanced accuracy.

6 **API Development & Docker Deployment**
Creating a FastAPI microservice and containerizing for scalable integration.

Gantt Chart:

Our project's detailed timeline outlines the key phases for developing the AI Fabric Classification module, from data preparation to final deployment.





Individual Component

Environmental Impact Prediction Module

Student: **Siriwardhana N.S.T.N**

Student ID: **IT22215338**

Project ID: **R26-IT-81**

Background

Textile manufacturing consumes large amounts of:

- Water
- Energy
- Raw materials

Producing new fabrics results in:

- High carbon emissions
- Large water consumption
- Environmental degradation



Reusing textile waste can significantly reduce these environmental impacts.

However, most waste trading platforms **do not provide environmental impact measurements**

Research Problem

Manufacturers and buyers cannot clearly measure the environmental benefits of reusing textile waste.

Key issues include:

> No automated calculation of CO_2 emission savings

> No estimation of water consumption reduction

> No sustainability scoring mechanism

As a result, organizations lack crucial data to evaluate the true environmental benefits.



Research Gap

Feature	Existing Solutions	Proposed Module
Environmental Metrics	Not available	CO_2 and water savings
Calculation Method	Manual LCA analysis	LCA dataset + ML regression
Sustainability Score	Not available	Green Score (A–F / 1–100)

Research Objectives

Main Objective

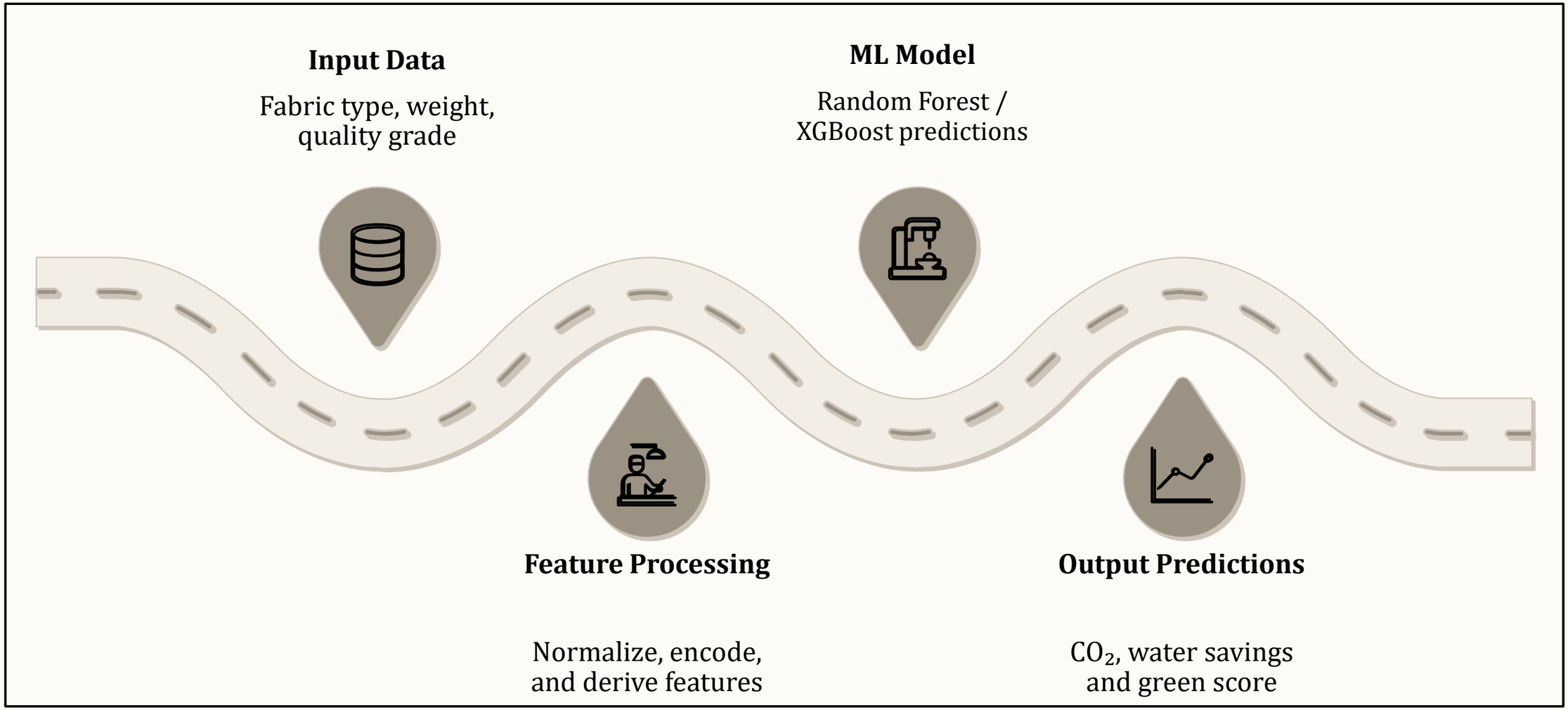
Develop a machine learning model that predicts environmental benefits from textile waste reuse, empowering sustainable choices.

Sub Objectives

- Collect Life Cycle Assessment (LCA) environmental data
- Engineer features such as fabric type, weight, and quality
- Train regression models to predict environmental savings
- Deploy the model as a FastAPI microservice

System Architecture

This module processes raw data to quantify the environmental impact of textile waste reuse, providing clear and actionable sustainability metrics.



Tools and Technologies

Our Environmental Impact Prediction Module relies on a specialized set of tools and data sources to accurately quantify sustainability benefits.



Python 3.9+

The core language for data processing, machine learning model development, and API implementation.



ML Libraries

Scikit-learn & XGBoost: Powerful frameworks for building and training robust regression models to predict CO_2 and water savings.



Deployment

FastAPI & Docker: Enabling high-performance API services and ensuring consistent, portable deployment of our environmental prediction model.



Key Datasets

LCA & Higg MSI: Foundational datasets providing crucial environmental impact metrics for accurate model training and validation.



Software Requirements

For the Environmental Impact Prediction Module

1

Functional Requirements

- Calculate environmental impact savings
- Predict CO_2 emission reduction
- Predict water usage reduction
- Generate sustainability Green Score

2

Non-Functional Requirements

- Prediction accuracy $R^2 > 0.85$
- Response time < 2 seconds
- Scalable API service



Work Breakdown Structure

This outlines the systematic approach for developing the Environmental Impact Prediction Module, from research to deployment.

Literature Review
Comprehensive study of existing methodologies for quantifying environmental impact and relevant Life Cycle Assessment (LCA) studies in textiles.

Data Collection & Curation
Sourcing and compiling Life Cycle Assessment (LCA) datasets specifically for textile materials and manufacturing processes.

Feature Engineering
Transforming raw LCA data and textile properties (type, weight, quality grade) into a clean, model-ready dataset for training.

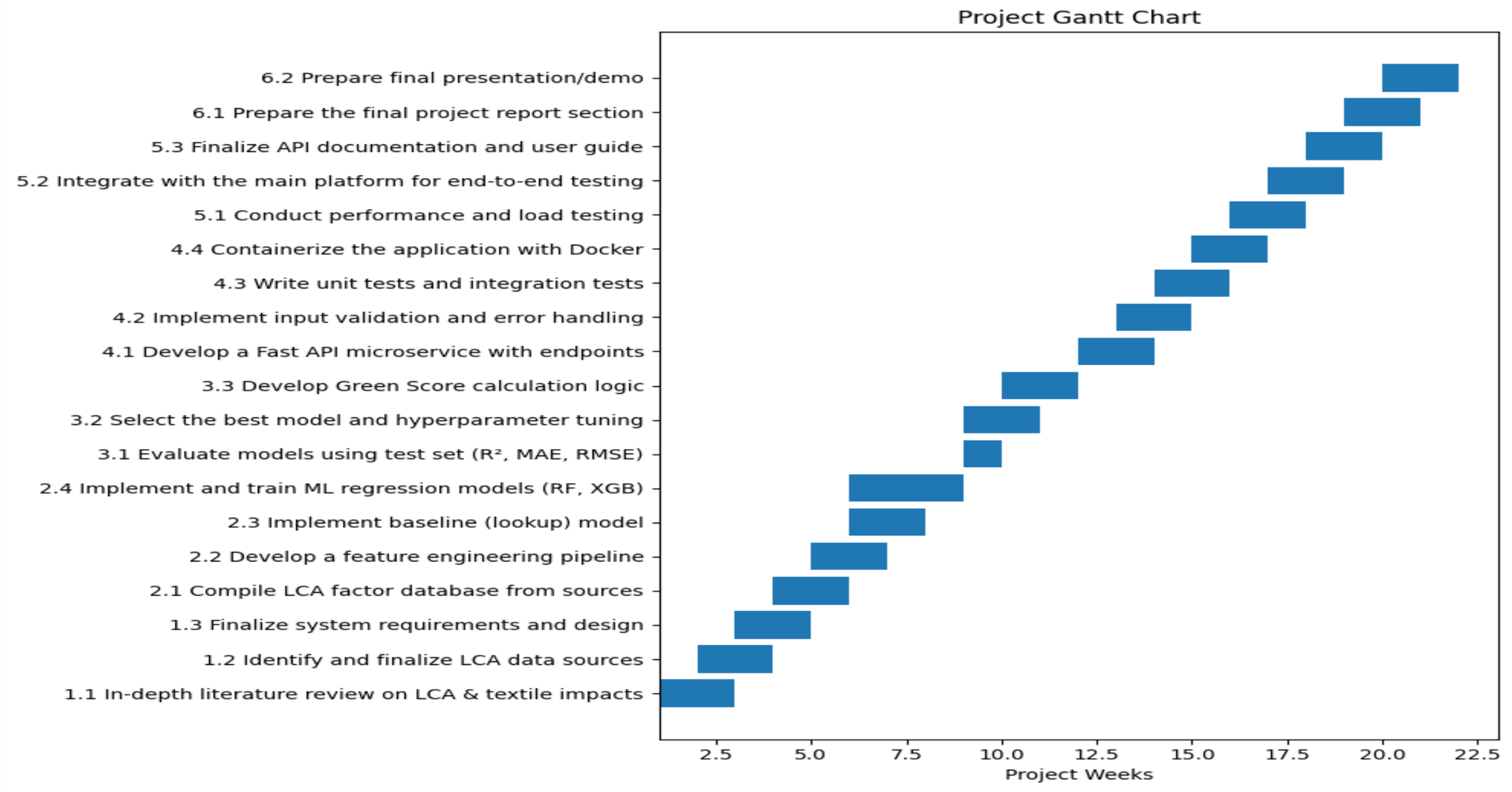
Model Training
Developing and training robust regression models, such as Random Forest and XGBoost, to accurately predict CO_2 and water savings.

Evaluation & Optimization
Rigorous testing of model performance metrics (e.g., R^2, RMSE), fine-tuning hyperparameters, and ensuring predictive accuracy and reliability.

API Development & Deployment
Building a high-performance FastAPI microservice and containerizing it with Docker for scalable and consistent deployment.

Gantt Chart: Weeks

Our detailed timeline for the Environmental Impact Prediction Module ensures a structured approach from data acquisition to final deployment.





Individual Component

Traceability, Recommendation & AI Pricing Module

Student: **Madushanka H.M.T.G**

Student ID: **IT22221278**

Project ID: **R26-IT-81**

Background

In textile waste trading, transparency and pricing accuracy are critical.

Current waste trading processes suffer from:

- Lack of traceability
- Manual pricing decisions
- Limited buyer-seller matching

Integrating AI pricing, recommendation systems, and traceability can significantly improve efficiency and trust.



Research Problem

Existing platforms lack:

- Digital traceability of textile waste origin
- AI-driven pricing models
- Intelligent recommendation systems for buyers

These limitations lead to:

- Unfair pricing
- Lack of trust between buyers and sellers
- Inefficient waste trading processes



Research Gap

Existing platforms struggle with crucial aspects of textile waste trading, highlighting a clear need for advanced solutions.

Feature	Existing Platforms	Proposed Module
Traceability	None	Digital material passport
Pricing	Manual estimation	AI regression pricing
Recommendations	Not available	AI recommendation engine

Our module directly addresses these limitations by integrating robust digital solutions.

Research Objectives

Main Objective

Develop a system that provides traceability, intelligent pricing, and recommendations for textile waste trading.

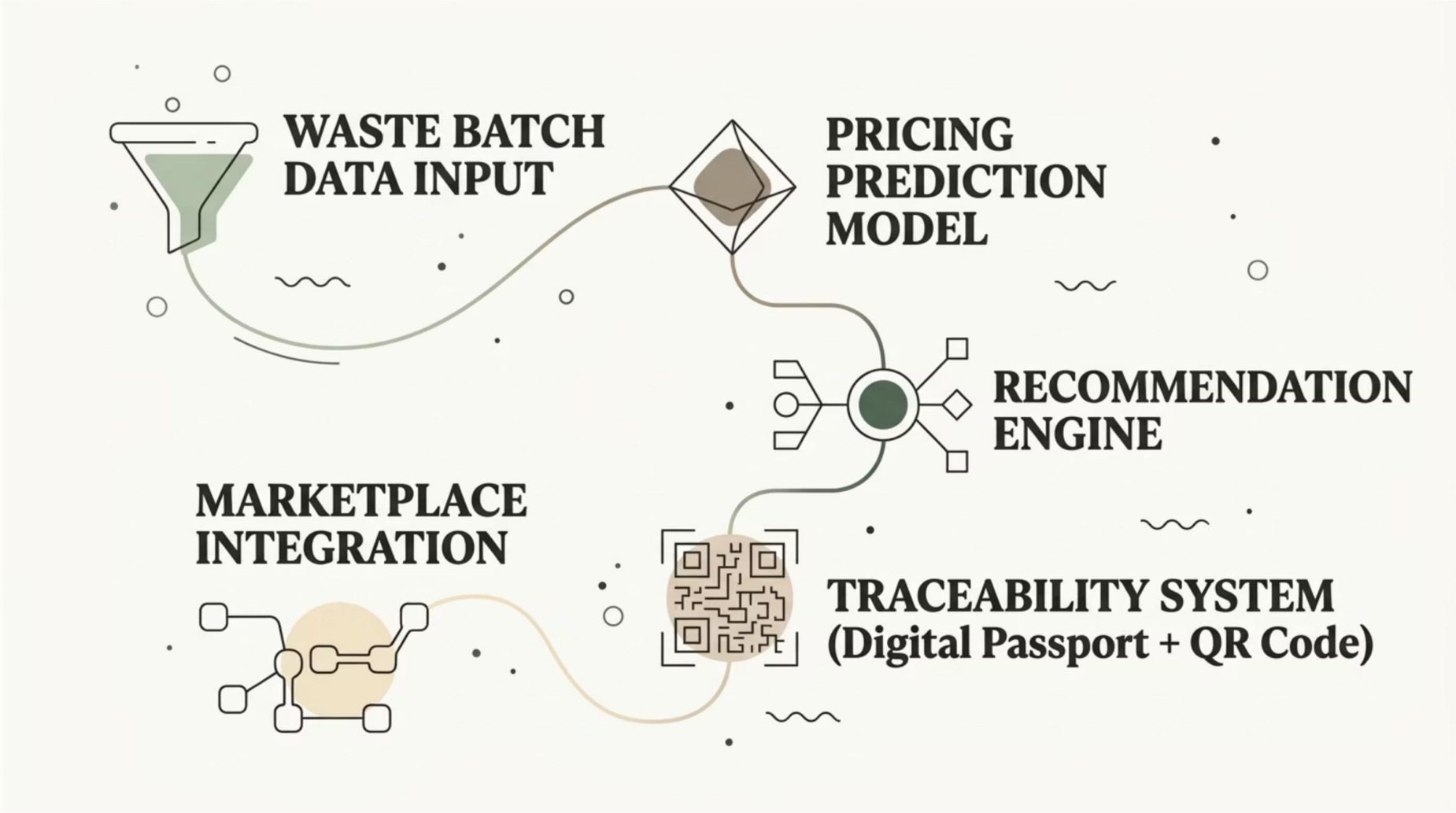
Sub-Objectives

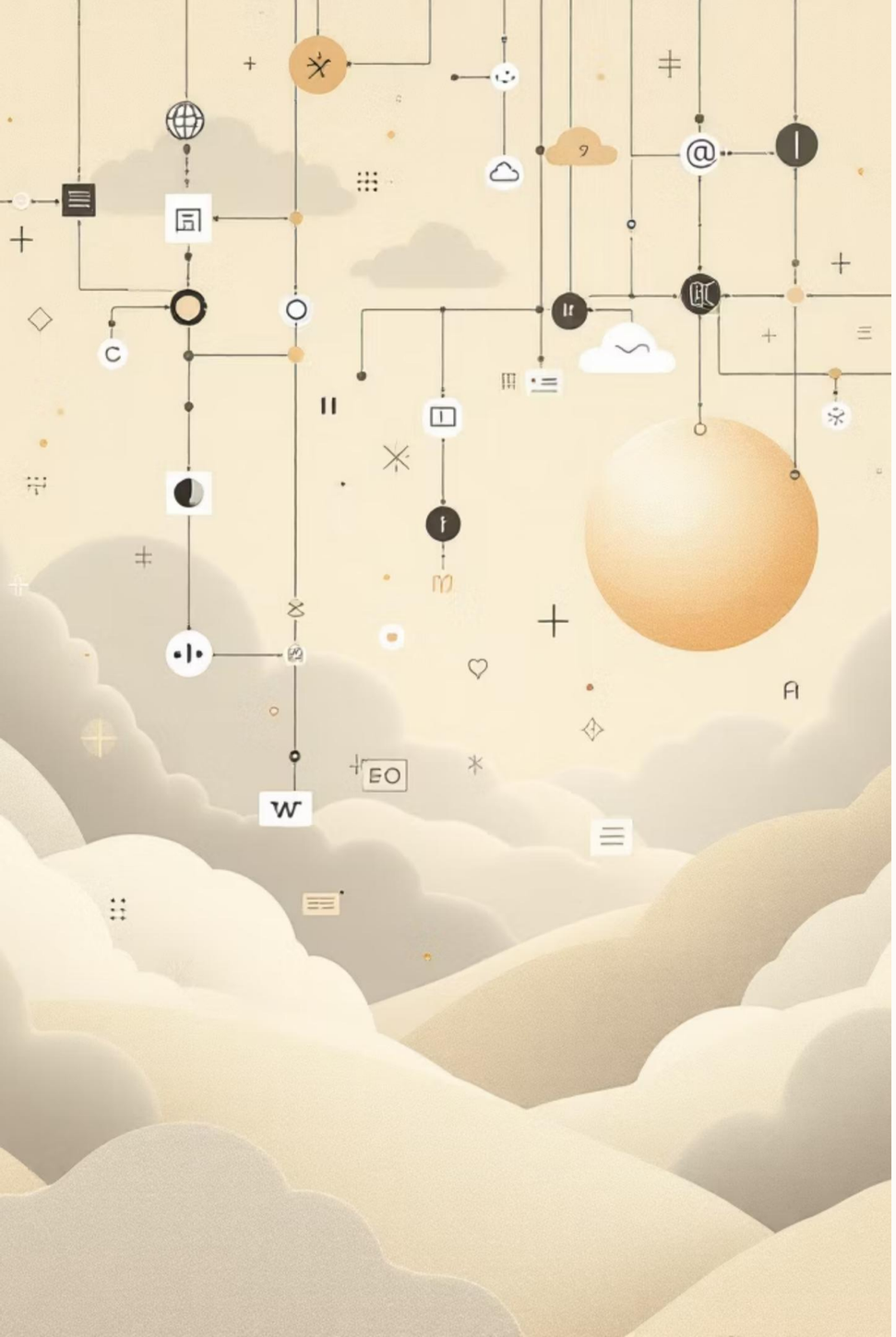
- Create a digital material passport for waste batches
- Develop an AI-based pricing prediction model
- Implement a recommendation system for buyers
- Deploy services through FastAPI APIs



System Architecture

This module integrates advanced AI with traceability to optimize the textile waste trading process, ensuring fair pricing and efficient matching.





Tools and Technologies

For the Traceability, Recommendation & AI Pricing Module

Python

The primary programming language, serving as the backbone for backend logic, A I model development, and data processing.

Scikit-learn

Utilized for developing robust machine learning models for AI pricing prediction and recommendation algorithms.

XGBoost

A powerful gradient boosting library employed for enhanced accuracy and performance in complex predictive tasks.

FastAPI

The high-performance web framework for building efficient and scalable API services for our module.

QR Code Generation

Enables the creation of digital material passports for textile waste batches, ensuring full traceability.

Docker Deployment

Containerization technology used to ensure consistent, portable, and scalable deployment of all module services.

Software Requirements

Our module's design is guided by specific functional and non-functional requirements to ensure optimal performance and user value.

Functional Requirements

- Predict resale price
- Generate buyer recommendations
- Provide traceability passport
- Generate QR code for batch tracking

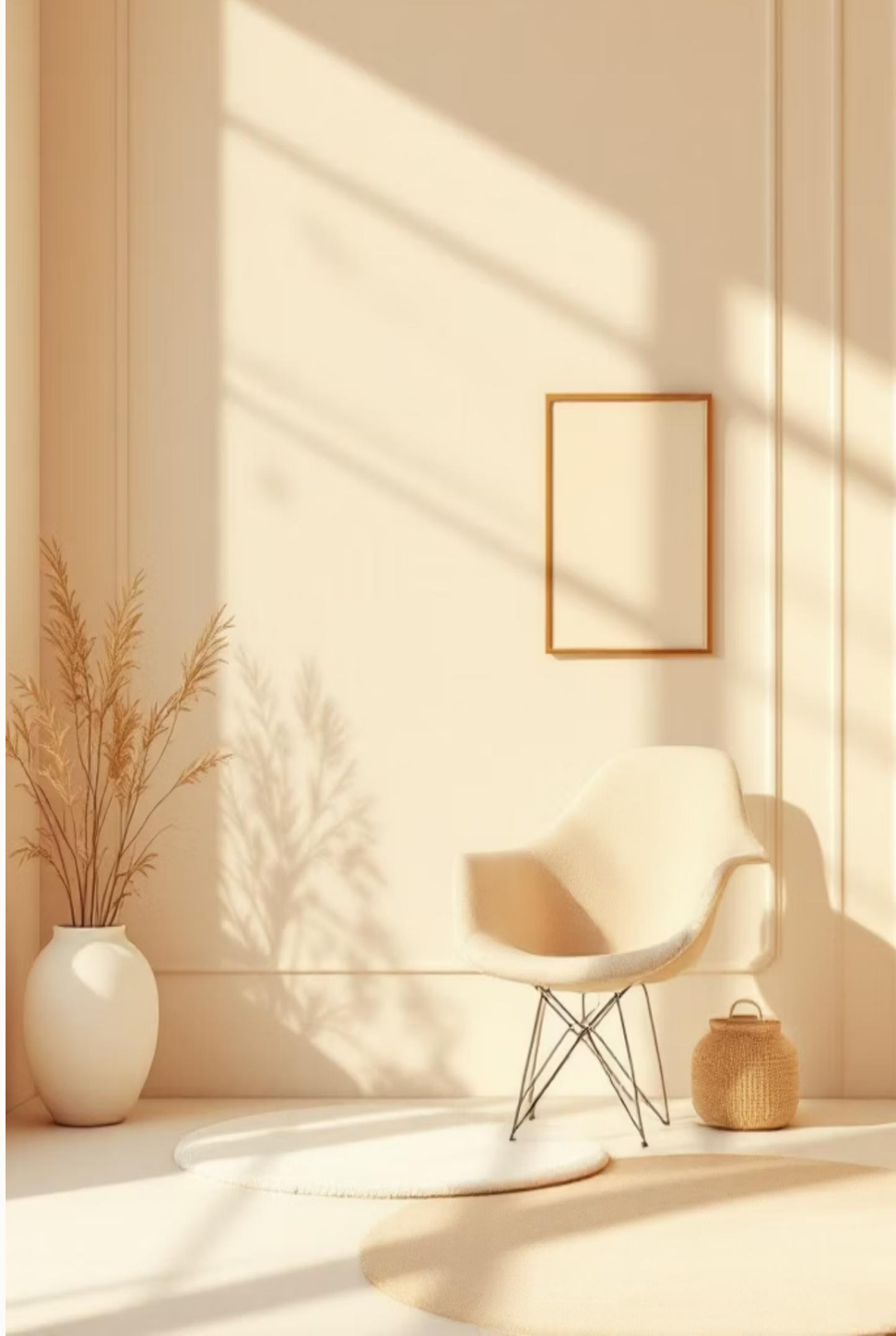
Non-Functional Requirements

- Secure and transparent data handling
- Accurate price prediction
- High system reliability

Work Breakdown Structure

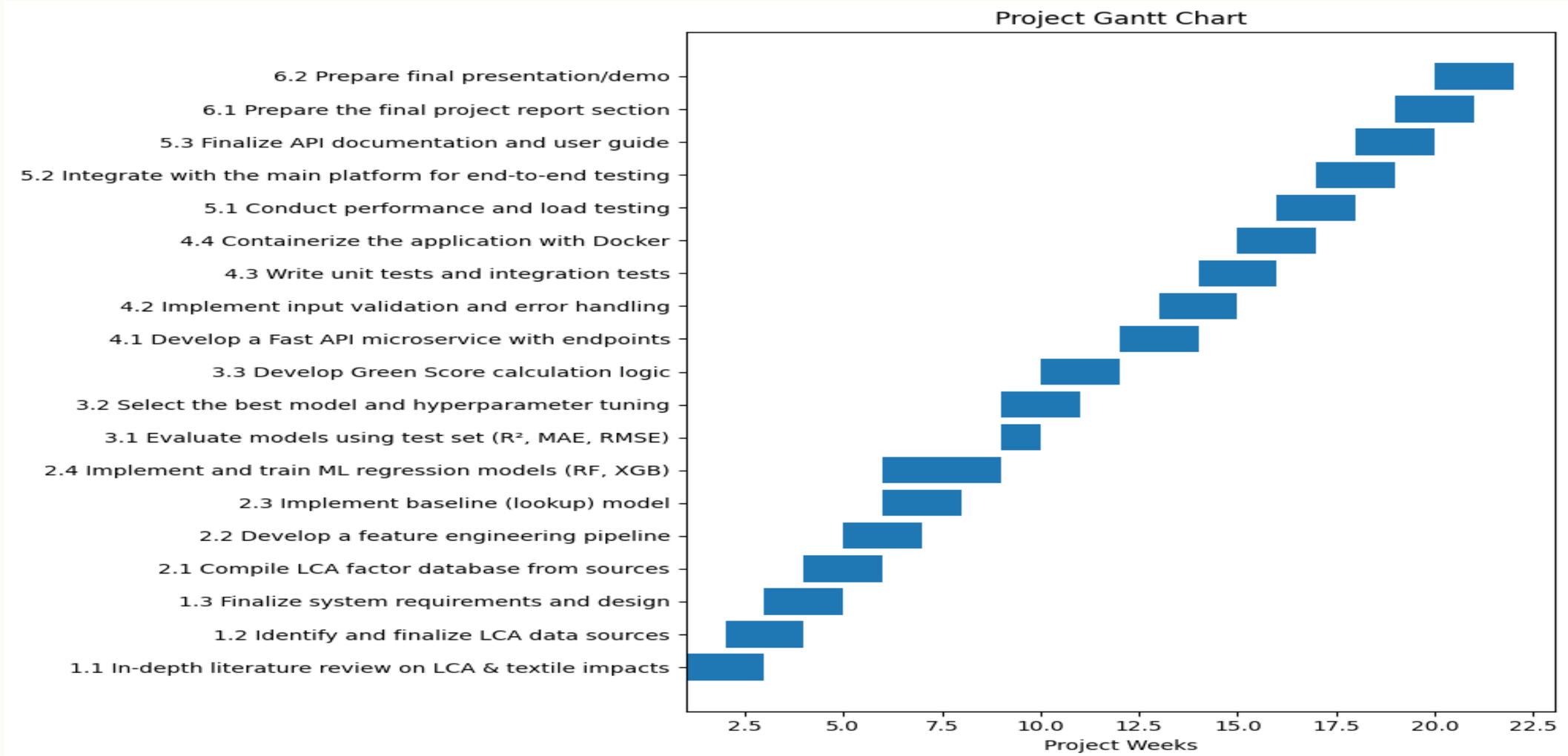
Our project's Work Breakdown Structure clearly outlines the key phases and deliverables, ensuring a systematic and manageable approach to development.

1. **Dataset Collection**
   Gathering and curating all necessary data for model training and system functionality.
2. **Feature Engineering**
   Transforming raw data into meaningful features to enhance model performance.
3. **Pricing Model Training**
   Developing and optimizing the AI-driven pricing prediction model.
4. **Recommendation Engine Development**
   Building the AI-powered system for intelligent buyer-seller matching.
5. **Traceability System Development**
   Implementing the digital material passport and QR code tracking.
6. **API Integration & Deployment**
   Connecting all components via FastAPI and deploying the module for production.



Gantt Chart: Weeks 4–22

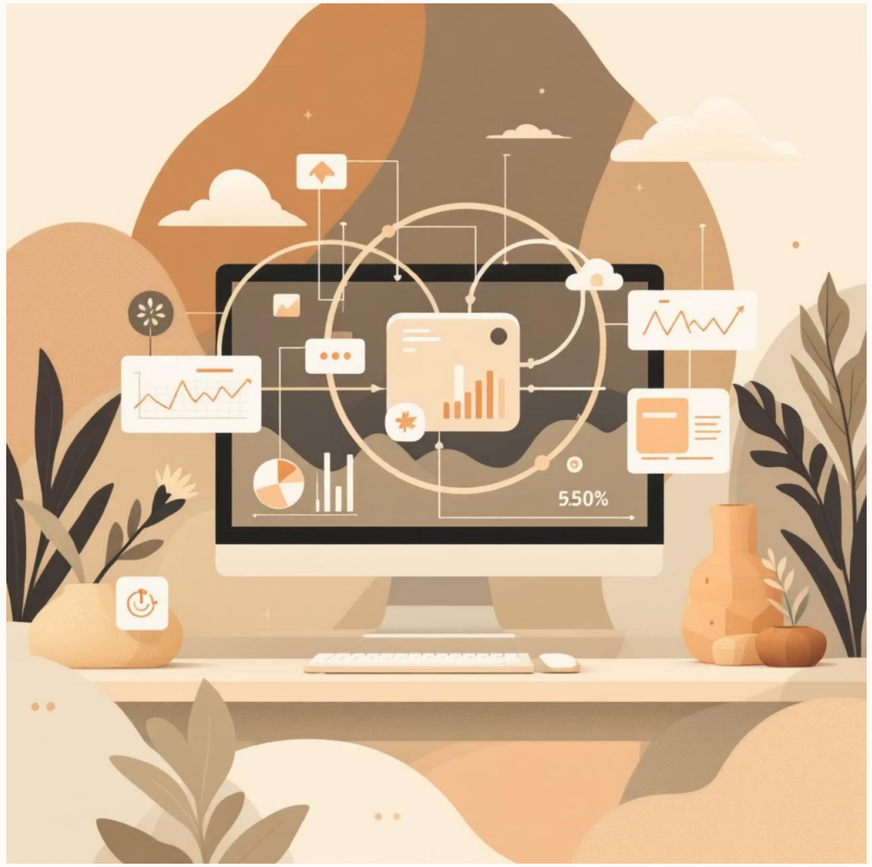
This detailed Gantt chart outlines the development timeline for the Traceability, Recommendation & AI Pricing Module, from initial data collection to final deployment.



Individual Component

Marketplace Platform & Predictive Analytics Module

Student: **Thalagala T.A.D.V**Student ID: **IT22281678**Project ID: **R26-IT-081**



Background

There is currently **no centralized digital platform** dedicated to textile waste trading in Sri Lanka.

A digital marketplace can:

- Connect manufacturers and buyers
- Improve waste reuse efficiency
- Provide environmental impact information
- Support circular economy practices.





Research Problem

Existing marketplaces:

- Are not specialized for textile waste
- Do not integrate AI models
- Do not provide predictive analytics for waste generation

This limits the ability of stakeholders to **plan sustainable waste management strategies.**

Research Gap

While progress has been made, current platforms fall short in addressing the specialized needs of textile waste management. Our module aims to bridge these critical gaps.

Feature	Existing Platforms	Proposed Module
Marketplace	Generic e-commerce	Specialized textile waste marketplace
Forecasting	None	ML waste prediction
Dashboard	Limited analytics	Interactive analytics dashboard

This table highlights how our specialized module offers advanced capabilities precisely where current solutions are lacking, focusing on intelligent features crucial for a circular economy.



Research Objectives

Main-Objectives

Our main objective is to develop an integrated digital marketplace platform with Predictive analytics capabilities for upcycling Sri Lanka's textile waste.

Sub-Objectives

1

Develop React-based marketplace interface
Create a user-friendly and responsive front-end for seamless interaction between buyers and sellers.

2

Build waste forecasting machine learning models
Implement AI models to accurately predict future textile waste generation and availability.

3

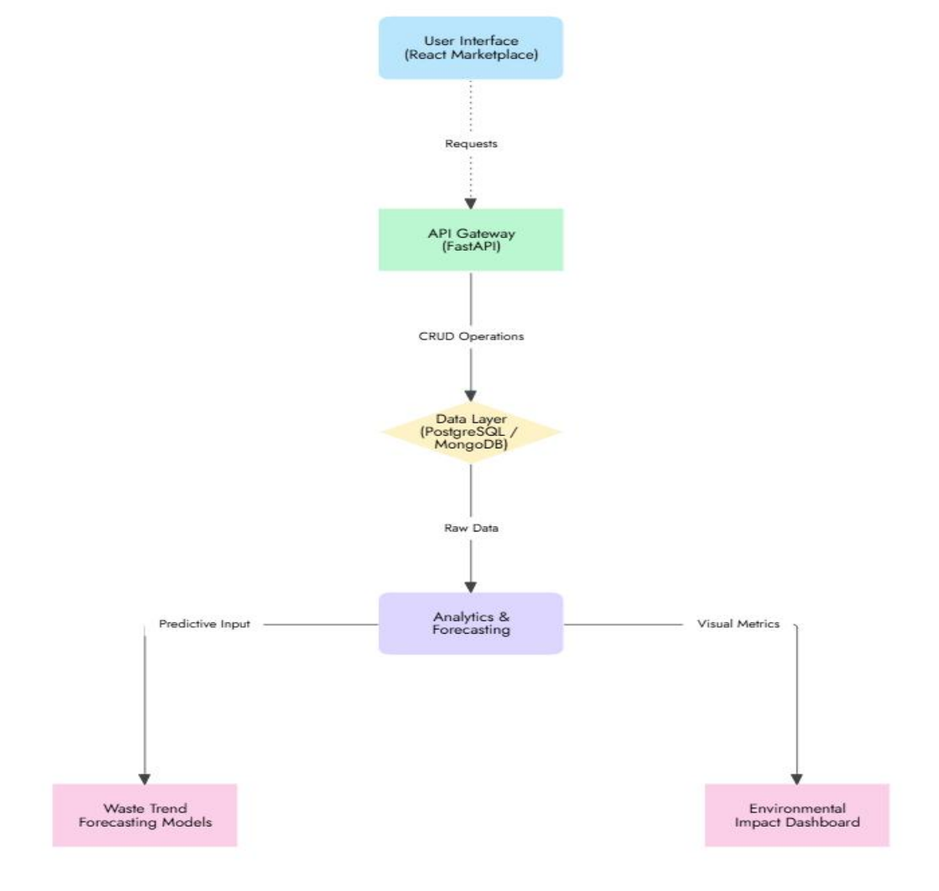
Create analytics dashboard for trends
Provide an interactive dashboard displaying key insights and trends in textile waste data.

4

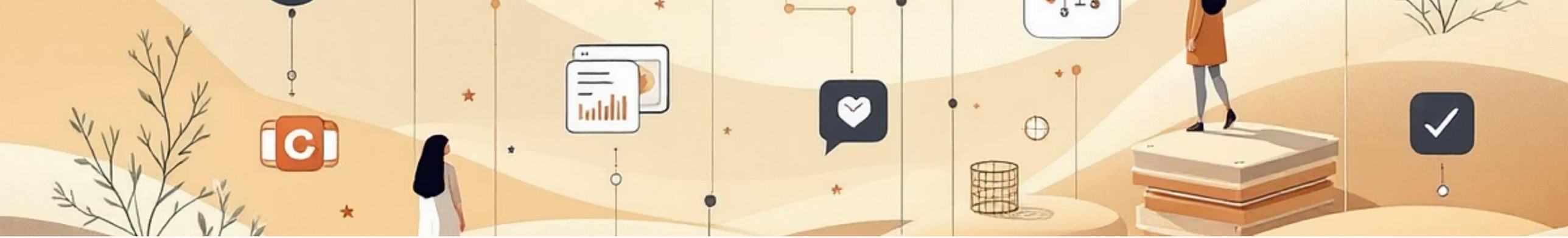
Integrate all AI modules into the platform
Ensure smooth and efficient communication between the various AI components and the core marketplace system.

System Architecture

Our system architecture is designed for scalability and efficiency, connecting user interfaces with intelligent backend services through a robust API gateway.



This modular approach allows for independent development and deployment of each component, ensuring system resilience and ease of maintenance.



Tools and Technologies

Our platform leverages a robust and modern technology stack to ensure scalability, efficiency, and seamless integration across all components.

Frontend

- React.js
- Tailwind CSS

Backend

- FastAPI

Databases

- MongoDB
- PostgreSQL

Machine Learning

- Scikit-learn
- XGBoost

Deployment

- Docker

Software Requirements

Our platform is designed with a clear set of functional and non-functional requirements to ensure a robust, user-friendly, and scalable solution for textile waste management.

Functional Requirements

- **User registration:** Secure profiles for buyers and sellers.
- **Waste listing:** Detailed descriptions and categorization of textile waste batches.
- **Search and filtering:** Advanced options for efficient discovery of materials.
- **Environmental impact visualization:** Data-driven insights on sustainability metrics.
- **Analytics dashboard:** Interactive charts and reports for market trends and performance.

Non-Functional Requirements

- **Scalable architecture:** Designed to handle increasing user and data loads.
- **User-friendly interface:** Intuitive design for ease of use across all functionalities.
- **High availability:** Ensures continuous access and minimal downtime.
- **Secure and transparent data handling:** Protects sensitive information and builds trust.
- **Accurate price prediction:** Reliable AI models for fair market value.
- **High system reliability:** Consistent performance and data integrity.

Work Breakdown Structure

Our project execution is meticulously planned through a Work Breakdown Structure, detailing each major phase from initial analysis to final deployment.

Requirement Analysis

Thoroughly defining the functional and non-functional requirements for the platform.

Forecasting Dataset Preparation

Collecting, cleaning, and preparing historical and real-time data for ML model training.

ML Model Development

Designing, training, and validating machine learning models for waste forecasting and pricing.

Dashboard Design

Creating an intuitive and interactive analytics dashboard for visualizing key metrics.

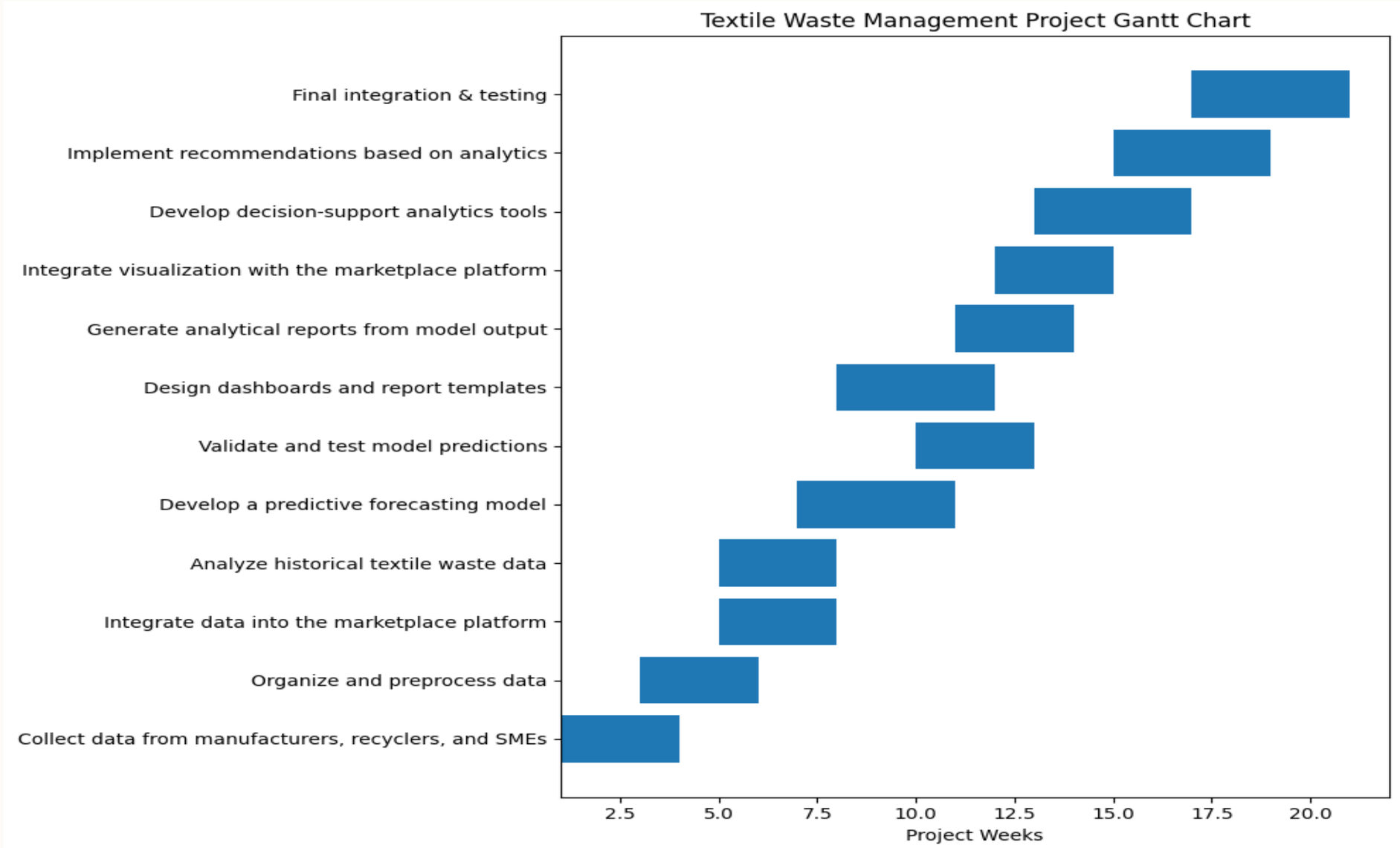
Frontend Development

Building the user-facing marketplace interface with React.js and Tailwind CSS.

API Integration and Deployment

Connecting all components via FastAPI and deploying the entire system using Docker.

Gantt Chart: Weeks



Conclusion

This project proposes an AI-powered digital marketplace for upcycling textiles waste in Sri Lanka, offering a comprehensive solution to address the nation’s textile waste crisis.

Intelligent Classification

Automated identification and categorization of textile waste.

Environmental Impact Measurement

Quantifying the positive ecological contribution of upcycling.

Transparent Traceability

End-to-end visibility of waste materials through the supply chain.

Data-Driven Insights

Optimized pricing and personalized recommendations for users.

Sustainable Trading Platform

A dedicated marketplace fostering circular economy practices.

Ultimately, this system promotes robust circular economy practices and significant environmental sustainability for Sri Lanka's textile industry.



Thank You!

AI-Powered Digital Marketplace for Upcycling Sri Lanka's Textile Waste

R26-IT-081